

Exercise 3.1 (DPI for tvd) Let $\mathcal{X}$ and $\mathcal{Y}$ be finite sets. Let $P_X, Q_X \in \mathcal{P}(\mathcal{X})$ and let $\{T(y|x)\}_{y \in \mathcal{Y}, x \in \mathcal{X}}$ be some conditional pmf. Suppose

$$P_Y(y) = \sum_{x \in \mathcal{X}} T(y|x) \cdot P_X(x) \quad \text{and} \quad Q_Y(y) = \sum_{x \in \mathcal{X}} T(y|x) \cdot Q_X(x)$$

for each $y \in \mathcal{Y}$. Prove that

$$\delta_{\text{tvd}}(P_Y, Q_Y) \leq \delta_{\text{tvd}}(P_X, Q_X).$$

Solution:

$$\begin{aligned} \delta_{\text{tvd}}(P_Y, Q_Y) &= \frac{1}{2} \cdot \sum_{y \in \mathcal{Y}} |P_Y(y) - Q_Y(y)| \\ &= \frac{1}{2} \cdot \sum_{y \in \mathcal{Y}} \left| \sum_{x \in \mathcal{X}} [T(y|x) \cdot (P_X(x) - Q_X(x))] \right| \\ &\leq \frac{1}{2} \cdot \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} |T(y|x) \cdot (P_X(x) - Q_X(x))| \\ &= \frac{1}{2} \cdot \sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} T(y|x) \cdot |P_X(x) - Q_X(x)| \\ &= \frac{1}{2} \cdot \sum_{x \in \mathcal{X}} |P_X(x) - Q_X(x)| = \delta_{\text{tvd}}(P_X, Q_X). \end{aligned}$$

Exercise 3.2 Consider the three pmfs P and Q , and S given by the probability vectors $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$, $q = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ and $s = (0, \frac{1}{2}, \frac{1}{2})$, respectively.

- Compute the unbiased (symmetric) error probabilities $\epsilon_{\text{sym},1}^*$ for binary hypothesis testing for all three pairs: (P, Q) , (P, S) and (Q, S) .
- Consider the problem of ternary hypothesis testing between the three distributions, when all three have equal priors. Find the minimal error probability and an optimal test.
Hint: Note that there are now more than two types of errors and there is no closed formula for the minimal error probability. You have to construct the optimal test manually, for example by trial and error.
- Compute $D(P\|Q)$ and $D(P\|S)$.
- Consider asymmetric hypothesis testing for P and S . For each $\epsilon > 0$, find $N_0(\epsilon) \in \mathbb{N}$ such that $\beta_n(\epsilon) = 0$ for all $n \geq N_0(\epsilon)$. What does that say about the limit $\lim_{n \rightarrow \infty} -\frac{1}{n} \log \beta_n(\epsilon)$? Interpret your result in item c) in this light.

Solution:

a.) We have

$$\delta_{\text{tvd}}(P, Q) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P(x) - Q(x)| = \frac{1}{6}, \quad \delta_{\text{tvd}}(P, S) = \frac{1}{2}, \quad \delta_{\text{tvd}}(Q, S) = \frac{1}{3}.$$

For pmf pair P and Q :

$$\epsilon_{\text{sym},1}^* = \frac{1}{2}(1 - \delta_{\text{td}}(P, Q)) = \frac{5}{12}.$$

For pmf pair P and S :

$$\epsilon_{\text{sym},1}^* = \frac{1}{2}(1 - \delta_{\text{td}}(P, S)) = \frac{1}{4}.$$

For pmf pair Q and S :

$$\epsilon_{\text{sym},1}^* = \frac{1}{2}(1 - \delta_{\text{td}}(Q, S)) = \frac{1}{3}.$$

- b.) We partition $\mathcal{X}$ into three (possibly empty) sets $\mathcal{A}_P$, $\mathcal{A}_Q$, and $\mathcal{A}_S$, such that we output $i \in \{P, Q, S\}$ if the sample x satisfies $x \in \mathcal{A}_i$. The partition of $\mathcal{X}$ needs to be optimal with respect to the probability of error:

$$\epsilon_{3,\text{sym},1}^* = \frac{1}{3} \min_{\mathcal{A}_P, \mathcal{A}_S \subset \mathcal{X}, \mathcal{A}_P \cap \mathcal{A}_S = \emptyset} (P(\mathcal{A}_P^c) + S(\mathcal{A}_S^c) + Q(\mathcal{A}_P \cup \mathcal{A}_S)).$$

Here we used that $\mathcal{A}_Q = \mathcal{X} \setminus (\mathcal{A}_P \cup \mathcal{A}_S)$, and hence $\mathcal{A}_Q^c = \mathcal{A}_P \cup \mathcal{A}_S$, to eliminate one of the variables. A calculation shows that $\epsilon_{3,\text{sym},1}^* = 1/2$. The optimal test is as follows: if the observed data satisfies $x \in \{1\}$, we say that it is generated from P ; if $x \in \{2, 3\}$, we say that it is generated from S .

- c.) We obtain

$$D(P\|Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)} = \frac{1}{2} \log \frac{9}{8},$$

and

$$D(P\|S) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{S(x)} = \infty.$$

- d.) Let $\mathcal{X} = \{1, 2, 3\}$ and $\mathcal{A} = \{x^n \in \mathcal{X}^n : x^n \text{ contains symbol } 1\}$. Since $P(1) > S(1) = 0$, there exists $N_0(\epsilon) \in \mathbb{N}$ such that $P^n(\mathcal{A}) \geq 1 - \epsilon$ for $n \geq N_0(\epsilon)$. However, $S^n(\mathcal{A}) = 0$ and thus, $\beta_n(\epsilon) = \min_{\mathcal{A} \subset \mathcal{X}^n, P^n(\mathcal{A}) \geq 1 - \epsilon} S^n(\mathcal{A}) = 0$ and $\lim_{n \rightarrow \infty} -\frac{1}{n} \log \beta_n(\epsilon) = \infty$, which corresponds to $D(P\|S)$.

Exercise 3.3 (Empirical typical set) Let $P \in \mathcal{P}(\mathcal{X})$ be a pmf over some finite set $\mathcal{X}$. The empirical typical set (of length n and tolerance ϵ) w.r.t. P is defined as

$$\mathcal{A}_{\text{emp},\epsilon}^{(n)}(P) := \{x^n \in \mathcal{X}^n : \delta_{\text{td}}(P_{[x^n]}, P) \leq \epsilon\},$$

where $P_{[x^n]}$ is the empirical distribution induced by x^n .

Prove that, for any $\epsilon \in (0, 1]$,

$$\lim_{n \rightarrow \infty} P \left[X^n \in \mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_X) \right] = 1.$$

Solution: We first prove the statement for $|\mathcal{X}| = 2$. Without loss of generality, assume $\mathcal{X} = \{0, 1\}$. In this case, we have

$$\delta_{\text{td}}(P_{[x^n]}, P_X) \leq \epsilon \iff \left| \frac{\sum_{i=1}^n x_i}{n} - \mathbb{E}[X] \right| \leq \epsilon.$$

Hence,

$$\mathbb{P} \left[X^n \in \mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_X) \right] = \mathbb{P} \left[\left| \frac{\sum_{i=1}^n X_i}{n} - \mathbb{E}[X] \right| \leq \epsilon \right] \geq 1 - \frac{\text{Var} \frac{1}{n} \sum_{i=1}^n X_i}{\epsilon^2}$$

by Chebyshev's inequality. Since $\text{Var} \frac{1}{n} \sum_{i=1}^n X_i = \frac{1}{n} \text{Var} X \rightarrow 0$ as $n \rightarrow \infty$, the above tends to 1 as $n \rightarrow \infty$.

For the case when $|\mathcal{X}| > 2$, for each $x \in \mathcal{X}$ we consider the “partial” empirical typical sets, and an induced binary random variable Y_x defined as follows

$$\mathcal{A}_{\text{emp},\epsilon,x}^{(n)}(P_X) := \left\{ \mathbf{x}^n \in \mathcal{X}^n : |P_{[\mathbf{x}^n]}(x) - P_X(x)| \leq \frac{2\epsilon}{|\mathcal{X}|} \right\}$$

$$Y_x = \begin{cases} 0 & \text{if } X \neq x \\ 1 & \text{if } X = x \end{cases}$$

It is not difficult to check that

$$X^n \notin \mathcal{A}_{\text{emp},\epsilon,x}^{(n)}(P_X) \iff Y_x^n \notin \mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_{Y_x}).$$

Note that $\mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_X) \supset \bigcap_{x \in \mathcal{X}} \mathcal{A}_{\text{emp},\epsilon,x}^{(n)}(P_X)$. Thus,

$$\begin{aligned} \mathbb{P} \left[X^n \in \mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_X) \right] &\geq \mathbb{P} \left[X^n \in \mathcal{A}_{\text{emp},\epsilon,x}^{(n)}(P_X) \quad \forall x \in \mathcal{X} \right] \\ &\geq 1 - \sum_{x \in \mathcal{X}} \mathbb{P} \left[X^n \notin \mathcal{A}_{\text{emp},\epsilon,x}^{(n)}(P_X) \right] \\ &= 1 - \sum_{x \in \mathcal{X}} \mathbb{P} \left[Y_x^n \notin \mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_{Y_x}) \right] \end{aligned}$$

Since $\mathbb{P} \left[Y_x^n \notin \mathcal{A}_{\text{emp},\epsilon}^{(n)}(P_{Y_x}) \right] \rightarrow 0$ for each x , as shown in the binary case, and the set $\mathcal{X}$ is finite, we have finished our proof.

Exercise 3.4 (Pinsker's inequality) For two pmfs P and Q , we want to show that

$$\delta_{\text{tvd}}(P, Q) \leq \sqrt{\frac{1}{2} D(P \| Q)}.$$

a) Let $f(x) = p \log x + (1-p) \log(1-x)$. Show that for $x \in [q, p]$,

$$f'(x) \geq 4(p-x),$$

and, thus, $f(p) - f(q) \geq 2(p-q)^2$.

b) For two Bernoulli pmfs $\tilde{P}$ and $\tilde{Q}$ defined by p and q respectively, prove Pinsker's inequality

$$\delta_{\text{tvd}}(\tilde{P}, \tilde{Q}) \leq \sqrt{\frac{1}{2} D(\tilde{P} \| \tilde{Q})}.$$

c) Let P and Q be two general pmfs. Define $\mathcal{S} = \{x \in \mathcal{X} : P(x) \geq Q(x)\}$. Let $\tilde{P}$ and $\tilde{Q}$ be two Bernoulli pmfs defined by $p = \sum_{x \in \mathcal{S}} P(x)$ and $q = \sum_{x \in \mathcal{S}} Q(x)$ respectively. Show that

$$\delta_{\text{tvd}}(P, Q) = \delta_{\text{tvd}}(\tilde{P}, \tilde{Q}) \quad \text{and} \quad D(\tilde{P} \| \tilde{Q}) \leq D(P \| Q).$$

Solution:

a.) We show by direct calculation

$$f'(x) = \frac{p}{x} - \frac{1-p}{1-x} = \frac{p-x}{x(1-x)} \geq 4(p-x)$$

Therefore,

$$f(p) - f(q) = \int_q^p dx f'(x) \geq \int_q^p dx 4(p-x) = 2(p-q)^2.$$

Note that, when written explicitly, we obtain

$$p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q} \geq 2(p-q)^2.$$

b.) Without loss of generality, we assume $p \geq q > 0$. By direct calculation,

$$\begin{aligned} \delta_{\text{td}}(\tilde{P}, \tilde{Q}) &= \frac{1}{2}(p-q) - \frac{1}{2}((1-p) - (1-q)) = p-q, \\ D(\tilde{P} \parallel \tilde{Q}) &= p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}. \end{aligned}$$

Therefore,

$$\delta_{\text{td}}(\tilde{P}, \tilde{Q})^2 = (p-q)^2 \leq \frac{1}{2} \left(p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q} \right) = \frac{1}{2} D(\tilde{P} \parallel \tilde{Q}).$$

c.) We show by direct calculation

$$\begin{aligned} \delta_{\text{td}}(P, Q) &= \frac{1}{2} \sum_x |P(x) - Q(x)| = \frac{1}{2} \sum_{x \in \mathcal{S}} (P(x) - Q(x)) - \frac{1}{2} \sum_{x \notin \mathcal{S}} (P(x) - Q(x)) \\ &= \frac{1}{2} \left(\sum_{x \in \mathcal{S}} P(x) - \sum_{x \in \mathcal{S}} Q(x) \right) - \frac{1}{2} \left(\sum_{x \notin \mathcal{S}} P(x) - \sum_{x \notin \mathcal{S}} Q(x) \right) \\ &= \frac{1}{2}(p-q) - \frac{1}{2}((1-p) - (1-q)) = \delta_{\text{td}}(\tilde{P}, \tilde{Q}). \end{aligned}$$

The data processing inequality directly gives $D(\tilde{P} \parallel \tilde{Q}) \leq D(P \parallel Q)$. Combining everything, we obtain

$$\delta_{\text{td}}(P, Q) = \delta_{\text{td}}(\tilde{P}, \tilde{Q}) \leq \frac{1}{2} D(\tilde{P} \parallel \tilde{Q}) \leq \frac{1}{2} D(P \parallel Q).$$

Exercise 3.5 (Continuity of entropy) For $\epsilon \in [0, \frac{1}{2}]$ and two distributions $P, Q \in \mathcal{P}(\mathcal{X})$ on a finite set $\mathcal{X}$ such that $\delta_{\text{td}}(P, Q) \leq \epsilon$, show that

$$|H(X)_P - H(X)_Q| \leq h(\epsilon) + \epsilon \log |\mathcal{X}|.$$

where $h : t \mapsto -t \log t - (1-t) \log (1-t)$ is the binary entropy.

Hint: Construct a pair of joint random variables (X_1, X_2) with marginals $P_{X_1} = P_X$, $P_{X_2} = \tilde{P}_X$, and $P[X_1 = X_2]$ large. Use Fano's inequality.

Solution: Define the non-negative functions

$$\begin{aligned} f_{\min} &: x \mapsto \min\{P_X(x), \tilde{P}_X(x)\}, \\ f_+ &: x \mapsto P_X(x) - f_{\min}(x), \\ f_- &: x \mapsto \tilde{P}_X(x) - f_{\min}(x). \end{aligned}$$

Notice that $\sum_x f_+(x) = \sum_x f_-(x) = 1 - \sum_x f_{\min}(x)$. Denote this value by α . Let (X_1, X_2) be a pair of joint random variables (distributing on $\mathcal{X}^2$) with pmf

$$P_{X_1, X_2}(x_1, x_2) = f_{\min}(x_1) \cdot \delta_{x_1, x_2} + \alpha^{-1} \cdot f_+(x_1) \cdot f_-(x_2).$$

This is a valid pmf since: First, it is obviously non-negative; Second, its marginals match P_X and $\tilde{P}_X$:

$$\begin{aligned} P_{X_1}(x_1) &= f_{\min}(x_1) + f_+(x_1) \cdot \sum_{x_2} \alpha^{-1} \cdot f_-(x_2) = f_{\min}(x_1) + f_+(x_1) = P_X(x_1), \\ P_{X_2}(x_2) &= f_{\min}(x_2) + f_+(x_2) \cdot \sum_{x_1} \alpha^{-1} \cdot f_+(x_1) = f_{\min}(x_2) + f_-(x_2) = \tilde{P}_X(x_2). \end{aligned}$$

Notice that

$$\begin{aligned} \mathbb{P}[X_1 \neq X_2] &= \sum_{x_1 \neq x_2} P_{X_1, X_2}(x_1, x_2) = \sum_{x_1, x_2} \alpha^{-1} \cdot f_+(x_1) \cdot f_-(x_2) = \sum_x f_+(x) \\ &= \sum_{x: P_X(x) > \tilde{P}_X(x)} P_X(x) - \tilde{P}_X(x) = \frac{1}{2} \sum_{x \in \mathcal{X}} |P_X(x) - \tilde{P}_X(x)| = \delta_{\text{tvd}}(P_X, \tilde{P}_X) \leq \epsilon. \end{aligned}$$

Without loss of generality, we assume $H(X_1) \geq H(X_2)$. In this case, by Fano's inequality,

$$0 \leq H(X_1) - H(X_2) \leq H(X_1 X_2) - H(X_2) = H(X_1 | X_2) \leq h(\epsilon) + \epsilon \log(|\mathcal{X}| - 1).$$

Since $\log(|\mathcal{X}| - 1) \leq \log|\mathcal{X}|$, we have

$$|H(X_1) - H(X_2)| \leq h(\epsilon) + \epsilon \cdot \log|\mathcal{X}|.$$

Exercise 3.6 Here we show that the set of sequences of type $[x^n]$ satisfies

$$|[x^n]| \geq \frac{1}{|\mathcal{T}_n|} 2^{nH(X)},$$

where X is distributed according to $P_{[x^n]}$.

a) Let X^n be n i.i.d. random variables sampled according to $Q(x)$. Let $P_{[x^n]}$ be the empirical distribution of x^n . Show that

$$P[X^n = x^n] = 2^{-n(H(X)_{P_{[x^n]}} + D(P_{[x^n]} || Q))},$$

where $H(X)_{P_{[x^n]}}$ is evaluated for the pmf $P_{[x^n]}$.

b) Let $Q(x) = P_{[x^n]}(x)$ and X^n as above. For an arbitrary $[\hat{x}^n]$, we define

$$x_{\max} = \operatorname{argmax}_x P_{[\hat{x}^n]}(x) - P_{[x^n]}(x), \quad x_{\min} = \operatorname{argmin}_x P_{[\hat{x}^n]}(x) - P_{[x^n]}(x),$$

and construct a new type class $[\tilde{x}^n]$ such that

$$P_{[\tilde{x}^n]}(x) = \begin{cases} P_{[\hat{x}^n]}(x_{\max}) - \frac{1}{n}, & x = x_{\max} \\ P_{[\hat{x}^n]}(x_{\min}) + \frac{1}{n}, & x = x_{\min} \\ P_{[\hat{x}^n]}(x), & \text{otherwise} \end{cases}$$

Show that

$$\frac{P[X^n \in [\tilde{x}^n]]}{P[X^n \in [\hat{x}^n]]} = \frac{P_{[x^n]}(x_{\min}) P_{[\hat{x}^n]}(x_{\max})}{P_{[x^n]}(x_{\max}) (P_{[\hat{x}^n]}(x_{\min}) + \frac{1}{n})}.$$

c.) Show that $P[X^n \in [x^n]] = \max_{[\hat{x}^n]} P[X^n \in [\hat{x}^n]]$ and use this to derive the desired inequality.

Solution:

a.) We can directly write down

$$\begin{aligned} P[X^n = x^n] &= \prod_x Q(x)^{nP_{[x^n]}(x)} = \prod_x 2^{nP_{[x^n]}(x) \log Q(x)} = 2^{n \sum_x P_{[x^n]}(x) \log Q(x)} \\ &= 2^{-n \sum_x (P_{[x^n]}(x) \log(P_{[x^n]}(x)/Q(x)) - P_{[x^n]}(x) \log P_{[x^n]}(x))} = 2^{-n(H(x)_{P_{[x^n]}} + D(P_{[x^n]} \| Q))}. \end{aligned}$$

b.) By combinatorics,

$$|[x^n]| = \frac{n!}{\prod_x k_{[x^n]}(x)!}.$$

We then have

$$P[X^n \in [\hat{x}^n]] = \sum_{\hat{x}^n \in [\hat{x}^n]} P[X^n = \hat{x}^n] = \frac{n!}{\prod_x k_{[\hat{x}^n]}(x)!} \prod_x Q(x)^{k_{[\hat{x}^n]}(x)}.$$

By the definition of $[\tilde{x}]$ and the empirical distribution, $[\tilde{x}^n]$ is obtained by reducing $k_{[\hat{x}^n]}(x_{\max})$ by one, and increasing $k_{[\hat{x}^n]}(x_{\min})$ by one. We see that

$$\frac{P[X^n \in [\tilde{x}^n]]}{P[X^n \in [\hat{x}^n]]} = \frac{Q(x_{\min})k_{[\hat{x}^n]}(x_{\max})}{Q(x_{\max})(k_{[\hat{x}^n]}(x_{\min}) + 1)}.$$

c.) We replace $nQ(x)$ with $k_{[x^n]}(x)$ to obtain

$$\frac{P[X^n \in [\tilde{x}^n]]}{P[X^n \in [\hat{x}^n]]} = \frac{k_{[x^n]}(x_{\min})k_{[\hat{x}^n]}(x_{\max})}{k_{[x^n]}(x_{\max})(k_{[\hat{x}^n]}(x_{\min}) + 1)}.$$

When $[\hat{x}^n] \neq [x^n]$, we have $k_{[\hat{x}^n]}(x_{\max}) > k_{[x^n]}(x_{\max})$ and $k_{[\hat{x}^n]}(x_{\min}) < k_{[x^n]}(x_{\min})$. Thus

$$\frac{P[X^n \in [\tilde{x}^n]]}{P[X^n \in [\hat{x}^n]]} > 1,$$

which is equivalent to

$$P[X^n \in [\tilde{x}^n]] > P[X^n \in [\hat{x}^n]].$$

We can now proceed further. When $[\hat{x}^n] = [x^n]$, we have

$$\frac{P[X^n \in [\tilde{x}^n]]}{P[X^n \in [\hat{x}^n]]} \leq 1.$$

We thus conclude that the maximum is achieved by $[x^n]$, i.e.

$$P[X^n \in [x^n]] = \max_{[\hat{x}^n]} P[X^n \in [\hat{x}^n]].$$

We set $Q(x) = P_{[x^n]}(x)$, then

$$P[X^n \in [x^n]] = \sum_{x^n \in [x^n]} P[X^n = x^n] \leq |[x^n]| 2^{-nH(X)_{P_{[x^n]}}},$$

and

$$P[X^n \in [x^n]] = \max_{[x^n]} P[X^n \in [x^n]] \geq \frac{1}{|\mathcal{T}_n|} \sum_{[x^n]} P[X^n \in [x^n]] = \frac{1}{|\mathcal{T}_n|},$$

which is equivalent to

$$|[x^n]| \geq \frac{1}{|\mathcal{T}_n|} 2^{nH(X)_{P_{[x^n]}}}.$$